INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 5: Discussion

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5 Discussion

Chapter 4 reported the measurements taken on Arbitrum One. Here each research question and each objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. Unless noted otherwise, same-window statements concern the matched cohort of contracts and holdout statements the spender cohorts.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP order the same contracts in different ways. They do: the interval of their agreement excludes both zero and one (Table 4.5). The divergence does not come from ties or from the rule for missing contracts: EndorseRank leaves few contract pairs tied, and keeping the contracts missing from a graph as isolated nodes leaves the agreement unchanged (Section 4.4).

The divergence follows from how the two graphs are built. The endorsement graph records the latest approvals and has fewer than half of the edges of the transfer graph, which records every transfer (T. Do et al., 2023), and the two methods restart differently. PageRank places stationary mass according to the edges it is given (Langville & Meyer, 2006; Page et al., 1999), so a contract that receives many transfers from active senders but few approvals ranks high under AWP and modestly under EndorseRank, and the reverse holds for a contract that many owners approve without being a transfer hub. H1 is supported, largely by construction, since two walks over different graphs with different restarts were not expected to agree. In practice one score cannot stand in for the other: an operator who adopts EndorseRank for its speed gets a different ranking and not a faster approximation of AWP.

5.1.2 RQ2: Same-window alignment

RQ2 asks how closely each score agrees with contemporaneous allowance and transfer checks. Each score follows the edges it walks on: the walks over transfers agree most with the transfer family and much less with the allowance family, the walks over allowances agree most with the allowance family, and every interval on a score's own family excludes zero (Table 4.3).

Circularity bounds H2. Every proxy in a family is computed from the same events as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955) and not an external test. Part of it is built in: AWP, whose edges weigh each transfer by its dollar value, follows the summed value of inbound transfers almost as closely as their number (Table 8.7). Ties do not bound H2: EndorseRank leaves few pairs tied, and the isolated-node rule changes no coefficient (Table 4.5). In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. None of these comparisons was fixed in advance, and the paired differences between EndorseRank and AWP, which carry no claim, are in Appendix 8.5.

5.1.3 RQ3: Computational cost

RQ3 asks what each score costs to compute on the same sample. The cost follows the edge count: EndorseRank is the cheapest score in time, memory and iterations, and the two hybrids, which walk the transfer layer, take longer than AWP (Table 4.1). Assembly and iteration are both linear in $|E|$ (Langville & Meyer, 2006), so the time ratio tracks the edge ratio (Table 4.1), and the difference in iteration counts adds little.

The scaling tables qualify this result more than they extend it (Table 4.2). Each stage is a subsample of the cohort that keeps the edges incident to its contracts, so the stages never test a graph larger than the cohort graph (Section 4.2.1). The damping and value-slope checks leave the alignment pattern as it was (Tables 4.6 and 4.7). H3 is supported on the matched cohort and its subsamples, again largely by construction, since both scores run the same solver. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ4, RQ5 and Section 5.1.6.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 has two clauses. The first asks whether scores frozen at the end of March 2026 predict new owner–spender approvals and new transfer senders on the spender cohort over the following three months. Every score predicts both labels, and the edge type decides how well (Table 4.10). On new approval pairs the interior mixtures and the in-approve degree lead, followed by EndorseRank and the allowance layer, while the scores built on transfers alone stay lower. On new senders AWP leads, ahead of the transfer in-degree, and the scores built on allowances alone rank lowest. W1 repeats this pattern among the same scores (Tables 4.14 and 4.16). The evidence concerns spenders; Section 5.1.6 turns to the traders.

The second clause asks whether any PageRank beats the raw degree it smooths, and the two walks answer it differently (Tables 4.11 and 4.16). AWP ranks later new senders above the transfer in-degree in both windows, with intervals above zero, and its lead holds on the spenders that had received a transfer before the freeze, so the spenders it scores zero do not create it (Table 4.12). EndorseRank, with the uniform restarts of the analysis plan, ranks new approval pairs below the in-approve degree in both windows. Its variant with AWP's activity restarts ranks them above the degree in both windows (Section 4.5.4), which suggests that the restart rule decides whether the allowance walk beats its count, a hypothesis for a registered test and not a finding. Known results lead one to expect the count to be hard to beat: in growing networks new links attach preferentially to nodes that already have many (Barabási & Albert, 1999; Newman, 2001), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022), and where degree correlations are weak, as on the Web, PageRank is on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is a reweighted approval count (Section 5.3.2). This does not undo the first clause, since the counts are statistics of the same edges. All of these contrasts were computed after the labels were known. The labels lie outside the window but belong to the same constructs as the scores, and neither contains any element of profit or repayment. RQ4 therefore shows the temporal predictive validity

of each construct, with the raw degree as the baseline that a PageRank must beat, and leaves credit risk aside; the liquidation label of W1 (Section 3.7.2), on which Section 5.1.6 compares the edge types, comes closest to it.

5.1.5 RQ5: Coupling the two edge types

Both halves of RQ5 get a negative answer, each under its own rule. The second half asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan, it does not. On W0 C-PR is no worse than the allowance layer on new approval pairs but falls behind the transfer layer on new senders (Table 4.11), and in the same window the secondary rule fails on all three families (Section 4.3.1). The first half asks whether the walk improves on a walk over the transfers alone. An exploratory comparison on W0 said that it did on new approval pairs, but it was made after the labels were known. Rule B, registered before the study extracted the July–September logs, decides the question, and its answer is no: C-PR again beat the transfer layer on new approval pairs, but non-inferiority was shown neither on new senders nor on the traders' liquidation-free close rate, and the rule needed all three conditions (Section 4.6.2). Against AWP, the comparator of the first sensitivity analysis, C-PR falls behind on new senders by a wider margin, and against EndorseRank it is slightly above on new approval pairs in both windows. The ablation and the layer weights explain both halves. S-PR keeps the transfer walk but restarts it from the allowance walk; it reproduces the transfer walk on the flow label and loses most of the endorsement signal, since a teleportation vector derived from endorsements changes little when transfer edges decide where the walk goes. In C-PR the allowance edges carry part of the walk, so the allowance signal survives on the spender holdout: every interior mixture is slightly above the allowance layer on new approval pairs and below the transfer layer on new senders (Tables 4.10 and 4.11). Its layers carry dollar amounts without a bound, apart from the cap on allowances, and restart uniformly, whereas AWP, which bounds the weight of each transfer and restarts at active senders, ranks new senders better than C-PR in both windows. Other couplings were not tested (Section 5.9).

Both halves count as results. The rule for the second half was fixed before any coupled score existed, so λ could not have been tuned to it, and it limits an obvious next step: a convex mixture of the two edge types at the transition level gained less over the allowance layer on new approval pairs than it lost against the transfer layer on new senders. The rule for the first half was registered before the study extracted its labels and fails on two of its three conditions. W1 establishes something narrower: adding allowance edges to a transfer walk gains information about authorization that holds on new data, at a loss on inflow larger than the margin of the registration.

5.1.6 Allowance against transfer on outcomes built from neither edge type

Unlike the spender labels of RQ4 and RQ5, which favor the scores built on their own edge type, the GMX trader outcomes are built from neither edge type and were measured after the scores were frozen. The comparison of the two edge types on them (Section 3.7.3) adds no research question and is reported as Claim C6 (Section 4.6.3).

This comparison bears on counterparty risk, the use that motivated the thesis (Chapter 1). For a protocol that extends leverage or credit to a contract account, a counterparty that is often force-liquidated is a riskier one, and forced liquidation is the nearest outcome to default in these data. On later liquidation avoidance in W1, the primary window, the approval count at the freeze ranked traders above the transfer count by $\Delta\tau = +0.084$, but its Bonferroni 97.5% interval, $[-0.026, 0.188]$, includes zero, as does the interval stratified by the number of closes, so R1 is not met; in W0, the secondary window, the difference was positive. The transfer count also ranks this outcome (Table 4.14), and the transfer scores rank as well the outcomes that grow with activity, the number of profitable closes and the realized gain (Table 4.17): they record how much a contract trades.

Four limits bound this reading. First, R2 is not met: the like-for-like PageRank pair, EndorseRank with AWP's restarts against AWP, is not resolved in either window, and EndorseRank again falls below the approval count it smooths (Table 4.18). Second, the comparison rests on a few dozen approval-receiving traders per window, all of whom avoided liquidation, against about three in four

of the others (Table 4.19), and every interval is wide. Third, every trader is a contract, so account type cannot explain the pattern, but the role of a contract may: the recipients were authorized to spend other addresses' tokens and close more positions, which fits managed or pooled trading contracts that the data do not identify. Fourth, R3 applies: the same signal is lower on the share of closes that were profitable, so the construct it ranks is forced-liquidation avoidance, not trading success; its lower non-loss share is partly built in, because the pipeline counts a liquidated close as not a loss. Within these limits, on the outcome a margin or lending protocol screens for, the comparison did not separate allowance-edge from transfer-edge information at the count level in the primary window.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

Six findings outside the objectives also shape the conclusions. Four limit the claims: the value slopes chosen in the revision were checked only in the same window (Section 5.8.1); the subsample stages never ran on a graph larger than the cohort graph (Section 5.1.3); on the trader cohorts no score ranks the share of profitable closes with an interval above zero (Section 4.6.1); and EndorseRank ranked both spender labels slightly above the allowance layer of C-PR, so how approvals are weighted may matter (Section 5.8.1). Two add to the claims: every trader contract that had received an approval avoided liquidation in both windows, in small samples (Section 5.1.6), and EndorseRank matches a closed form for every depth-one spender (Section 5.3.2). That AWP beats its count while EndorseRank does not, and that the variant with activity restarts does, is not among them, because it belongs to the second clause of O4.

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ contains neither 0 nor 1	Table 4.5, last row: the interval $[0.163, 0.185]$ lies strictly inside $(0, 1)$	Met (largely by construction)
O2	A score agrees with a construct when its interval on that family excludes 0; agreement with the construct of a score's own edges is an intended-construct check	Table 4.3: every walk over transfers alone on the transfer family (0.572 to 0.659) and both walks over allowances on the allowance family (0.585 and 0.597); every interval excludes 0 except EndorseRank's on Sybil stability; the proxies come from the same events as the graphs	Met (largely by construction)
O3	Runtime ratio given alongside the edge ratio, under a single fixed protocol	Tables 4.1 and 4.2: mean, SD, memory, iterations, $\lvert V\rvert$, $\lvert E\rvert$ per configuration; time ratio 2.17 beside an edge ratio of 2.23; one solver, so time follows $\lvert E\rvert$	Met (largely by construction)
O4	A score predicts a label when its interval excludes 0; a PageRank improves on its degree only when the interval of the paired difference lies above 0	Tables 4.10, 4.14 and 4.16: every score predicts both labels in W0 and W1. Tables 4.11 and 4.16: AWP minus transfer in-degree $+0.043$ $[0.038, 0.048]$ (W0) and $+0.022$ $[0.017, 0.027]$ (W1); EndorseRank minus in-approve degree -0.019 $[-0.026, -0.011]$ (W0) and -0.034 $[-0.038, -0.029]$ (W1); all computed after the labels were known	Met (first clause); met for AWP, not for EndorseRank (second clause)
O5	(a) Rule A of 9 October 2026: on W0, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the leading layer's interval on the transfer and allowance families, and it exceeds both layers on Sybil stability. (b) Rule B, registered on 9 October 2026: on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs and is not worse by more than 0.01 (lower 95% bound above -0.01) on new senders and on the traders' liquidation-free close rate	(a) Table 4.11: no worse holds ($+0.029$ $[0.022, 0.036]$), better fails (-0.034 $[-0.040, -0.029]$). Same window: transfer (0.365) lies outside the interval of C-PR ($\lambda = 0$) and allowance (0.477) outside that of C-PR ($\lambda = 1$) (Table 4.3); Sybil stability lies between the layers (Table 4.4). All three parts fail. (b) Table 4.15: new approval pairs $+0.110$ $[0.098, 0.122]$ holds; new senders -0.028 $[-0.033, -0.023]$, wholly below -0.01, fails; liquidation-free close rate $+0.000$ $[-0.014, 0.010]$, lower bound below -0.01, fails	(a) Not met; (b) not met

5.3 Theoretical implications

The results bear on three areas of theory: propagation semantics on financial graphs, the construct validity of on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999). H.-D. Do and Do (2023) carried this reading over to Ethereum transactions, and AWP (T. Do et al., 2023) refined it with value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner who lets a spender act for them issues a directed endorsement of spending authority (Vogelsteller & Buterin, 2015). The RQ1 divergence shows that changing what the edges mean changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each valid relative to the relationships its edges encode. Peer-to-peer and web trust systems reached the same conclusion from the other side: in EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) the seed set and the edge definitions largely decide the outcome. EndorseRank's edge is native to a token standard, where theirs belonged to a message-passing overlay or a hyperlink graph.

The holdouts add a qualification that the propagation metaphor hides. Changing the edge type changed what the score could anticipate: on new approval pairs the scores built on allowance edges led and those built on transfers alone stayed lower, and on new senders the reverse held, in both windows. Whether propagating along the edges added to that depended on the side: AWP beat its count, EndorseRank with uniform restarts did not, and its variant with activity restarts did (Section 5.1.4). On these data the value lies in the choice of edge, and the restart rule appears to decide whether the walk improves on the count. When the two layers share one walk, the allowance signal survives where allowance edges carry the walk and is lost where they only seed its restarts (Section 5.1.5).

5.3.2 EndorseRank on depth-one spenders

The allowance graphs keep only the approvals of the five tokens that a cohort contract grants or receives (Section 3.6), so most nodes are pure owners, with out-edges only, or pure spenders, with in-edges only. A cohort contract can be both, so the graph also has paths of more than one step. For a spender whose owners receive no approvals, EndorseRank has a closed form.

Proposition. Let PageRank be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u \frac{w(u,v)}{o(u)}, \tag{5.1}$$

where $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all the spenders it approves.

Derivation. At the fixed point, $r_v = (1-d+d\,r_D)/|V| + d\sum_u r_u\,w(u,v)/o(u)$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1-d+d\,r_D)/|V|$, the constant c of Section 5.6.3. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On the graph at T_{obs}, 12,039 of the 14,406 cohort spenders have depth one, and EndorseRank as computed in this thesis lies within 1% of Equation (5.1) for every one of them; of the deeper spenders, which have an owner that itself receives approvals, 26.4% do. Over all spenders the share is 87.9%, and over the W0 spenders at t_1 it is 73.7%. Over the cohort spenders EndorseRank's Kendall τ with δ is 0.925, a point estimate (scripts `run_same_window.py` and `run_holdout.py`).

Two consequences follow. First, for a depth-one spender EndorseRank only reweights the approval count, counting an owner that approves many spenders for less at each; besides the time and value weights, that normalization is all that separates it from the count, which ranks new approval pairs better (Section 5.1.4). For these spenders the negative answer to the second clause of RQ4 for

EndorseRank thus has an analytic basis: there was no propagated information for the walk to add. Second, the graph does have deeper paths, and for most deeper spenders the score departs from the closed form, so multi-hop propagation is present in this graph; it did not make EndorseRank beat the raw count out of window. The ego network cuts longer paths (Section 5.8.3), so testing propagation further would need a deeper sample, such as the two-hop neighborhood of the cohort or a chain-wide allowance graph, or restarts seeded at vetted contracts.

5.3.3 Multi-construct domain alignment

The checks were designed to separate transfer centrality from allowance delegation, and each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs (Cronbach & Meehl, 1955), and the integrated view of Messick (1995) goes further, counting the consequences of using a score as part of its validity. Rank correlations (Kendall, 1938; Spearman, 1904) put this logic into practice for ordinal reputation scores without requiring classification thresholds, while the paired bootstrap (Efron, 1979) makes the pattern testable. The same theory says what such a pattern cannot settle: when two instruments measure different constructs, the one that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. Outcomes built from neither edge type are the exception, since neither instrument was built for them, which is why the thesis compares the edge types there (Section 5.1.6); even there, which edge type leads depends on the outcome.

5.3.4 Reputation under cheap pseudonyms

Identities are cheap to create on public ledgers (Nakamoto, 2008), and under Ethereum's account model a new address needs no registration step at all (Wood, 2025). Any reputation system that rewards inbound transfers or allowances is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls (Douceur, 2002). The empirical findings do not rule

this out. Section 5.6 formalizes the cost of attacking an allowance graph, where a closed farm of addresses that approve one another needs, unlike transfer wash trading, no token balance and pays little beyond gas.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers have to choose between transfer-hub detection and authorization-aware reputation. The measured results support choosing the score by scenario over recommending a single one.

5.4.1 Transfer-hub and flow monitoring

A protocol whose priority is describing current flow (for market-maker identification, routing analysis, or transfer-based compliance heuristics) can read that flow from in-degree and in-value directly. AWP (T. Do et al., 2023) adds propagation, time and value weights and restarts at active senders. It agrees with present inflow (Table 4.3), and out of window it ranked later new senders better than any other score, the raw transfer in-degree included, in both windows (Tables 4.10 and 4.16). A protocol that wants to anticipate flow has reason to use AWP over the count.

5.4.2 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most (token vaults, allowance-gated integrations, or risk engines that treat approvals as acts of trust), protocols have reason to use allowance edges. The cheapest form is the spender's in-approve degree, which predicted new approvals better than EndorseRank in both windows (Tables 4.11 and 4.16) and needs no solver at all. EndorseRank is its propagated version. Its allowance alignment (Table 4.3), its holdout result (Table 4.10) and its short solve time (Table 4.1) make it a reasonable choice where the operator wants a ranking that also reflects who the endorsing owners are. For most spenders, however, it reflects them only through how many spenders each owner approves (Section 5.3.2), a property not shown to improve any label, and under uniform teleportation the propagated score is open to cheap inflation by closed farms

(Section 5.6). The variant with activity restarts ranked new approval pairs above the count after the labels were known, so it is the propagated allowance score to test. In return for authorization semantics and speed, operators accept a score that carries little transfer information.

5.4.3 Evaluation design and combined use

Whether to combine the two signals is a question about deployment and about method, and the evidence answers the two differently. An operator can run EndorseRank for continuous authorization-aware monitoring and AWP for periodic transfer-hub audits, at the cost of both solves. Folding the two edge types into one walk did not pay on these data: C-PR costs more than AWP (Table 4.1) and trails AWP on new senders in both windows (Section 5.1.5), so an operator free to choose should keep the two scores apart. For an operator tied to a single transfer walk weighted by dollar amounts, adding allowance edges bought authorization information that held on new data, at a loss on inflow larger than 0.01, so rule B is not met. Because the pipeline is open, the analysis can be replicated on any ERC-20 chain that has allowance logs and a daily price series for its main tokens (Appendix 8.1); integration with live risk oracles is left for future work (Chapter 6).

5.4.4 Industry adoption scenarios

The scenarios below say for whom each score is useful and for whom it is not.

- DEX operators. A protocol that sets fee tiers or monitoring priority by user needs a score for its users, and its users are mostly externally owned accounts, which this thesis does not score (Section 3.2). The scores can instead rank the routers, vaults and pools that the DEX integrates with.
- Lending and margin protocols. Lending markets want to put more of their capital to use without cutting collateral (Bartoletti et al., 2021), and the stress of 2020 showed the cost of updating risk parameters late (Gudgeon, Perez, et al., 2020). When the counterparty is a contract, their screening question is whether it will be force-liquidated. On that outcome a lead of the approval count over the transfer count was resolved in W0 but not in the primary

window, the PageRank pair is unresolved, and the signal, lower on trading success, rests on a few dozen contract traders per window (Section 5.1.6). A screen built on it would need a registered test on a larger sample. The scores are not tools for predicting default.

- Wallet and security tooling. Users and integrators want to know whether a spender contract is widely authorized. A badge that shows how many owners have approved a spender, or its EndorseRank, makes that delegation explicit (Lundfall, 2020; Vogelsteller & Buterin, 2015); it conveys the position of a graph node and is neither a security audit nor a guarantee against malicious contracts.
- Liquidity providers and market makers. To control inventory and spreads, these participants need to find transfer hubs and see how flow is structured. Transfer counts and AWP are the suitable tools for that goal (T. Do et al., 2023), and their benefit is lower screening cost, not higher profit and loss.
- Airdrop and governance programs. Sybil clusters routinely abuse token distributions and governance eligibility (Douceur, 2002; Luo et al., 2025). EndorseRank as defined here offers little protection below the top of its ranking: a closed farm collects about three times its population share and can concentrate it on one target, so about 22 addresses lift a contract into the cohort's top tenth, although reaching the top would take more than a hundred thousand (Section 5.6). Programs can require diverse incoming endorsements from owners with independent histories, discount mass that circulates inside closed clusters, or send teleportation and dangling mass only to a vetted seed set; none of these stops an attacker who buys endorsements from honest owners.
- Analytics and compliance providers. Exchanges, funds and compliance teams need counterparty rankings that outsiders can recompute. EndorseRank and AWP can be published for defined windows and parameters and audited, unlike proprietary machine learning models (Packin & Lev-Aretz, 2024), provided that their Sybil exposure is disclosed and they are not presented as profit predictions.

In none of these scenarios does reputation replace collateral.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each of them can recompute it and see how it might be manipulated. EndorseRank and AWP can be recomputed from public logs and a public daily price series with the open pipeline (Appendix 8.1), and Sections 5.6 and 5.7 document how they can be manipulated.

5.5 Societal and practical impact

In the DeFi market, vetting a counterparty is expensive, and the capacity to do it is unevenly spread. Wealthy actors may be able to pay for proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. A score built from the permissions users have already granted, and cheap to recompute (Table 4.1), lowers the barrier to vetting the contracts that ask for those permissions. An approval that a user granted earlier is one route by which tokens are drained (Section 2.17). A spender-centered endorsement score therefore conveys information about a risk users face directly and that transfer volume cannot measure.

For protocol developers, the results argue for replacing a single "reputation" construct with specific scores, each fitted to the decision it informs, which reduces the risk of using a flow score to decide authorization.

Where the decentralized credit scoring systems discussed by Packin and Lev-Aretz (2024) are opaque, a score built from public events and public prices with versioned parameters supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed (Zetzsche et al., 2020). It also brings a duty, described in Section 5.7, to ensure that an ordinal, graph-based statistic is not mistaken for an identity, a credit history, or a security audit.

Any reputation metric that affects how value is allocated is open to exploitation; the next section models what it costs to manipulate EndorseRank.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack exploits cheap pseudonyms to create a false impression of trustworthiness (Douceur, 2002). On a transfer graph, a malicious actor can inflate in-degree and in-flow artificially by moving assets among accounts it controls, paying for this in transaction fees and possible inventory loss (Cong et al., 2023). Allowance graphs expose a different attack surface, since each edge records an authorization, not an actual transfer. In this section we formulate a simple cost model for inflating EndorseRank with fake allowances. It builds on the optimal link farms of Gyöngyi and Garcia-Molina (2005) and on the result of Cheng and Friedman (2005) that symmetric functions such as PageRank are not Sybil-proof (Section 2.6), and adds the cost of an allowance edge and the size of the gain on this study's graph.

5.6.1 Attack setup

Let the attacker control a set of addresses $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest addresses in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) of weight $w_{uv} > 0$ records that u has authorized v to spend its tokens. The weight is that of Equation (3.7): for each of the five tokens, the time weight of the latest in-window approval times the bounded value weight of its US-dollar amount, summed over tokens. With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of (Page et al., 1999):

$$\mathbf{r} = d\, P^{\top}\mathbf{r} + \frac{1-d}{|V|}\mathbf{1}, \tag{5.2}$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes (Langville & Meyer, 2006). The fixed point is the same as that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker's objective

is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} r_i, \tag{5.3}$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also buy a set $E_{\mathcal{H}\to\mathcal{S}}$ of edges from honest owners into $\mathcal{S}$. Costs are counted over q observation windows, the number of score refreshes for which the targets are meant to hold their rank. Gas is the only cost that every edge incurs. The other components arise only for particular edges or strategies.

1. Gas cost. Each `approve` or `permit` call consumes gas; write c_{gas} for the expected fee of one allowance update. EndorseRank reads only the `Approval` events inside its observation window, so an edge enters a window's graph only if its approval is emitted in that window. To keep $E_{\mathcal{S}}$ in place for q windows, the attacker has to emit every approval again in each window, which also keeps its time weight near the maximum $\sigma(0) = 0.86$, and the gas is paid q times:

$$C_{\text{gas}} = q\,|E_{\mathcal{S}}|\,c_{\text{gas}}.$$

2. Exposure cost. An approval locks no tokens, and the owner need not hold any balance of the token when it approves. The amount costs nothing either: an unlimited allowance takes the full value weight V without any balance behind it, and after row normalization a single out-edge gives $P_{uv} = 1$ whatever its weight. The list of five tokens does not raise this cost, since approving a listed token needs no balance either. An edge inside $\mathcal{S}$ therefore exposes nothing and costs nothing beyond gas. Exposure arises only when an attacker address that holds tokens approves a spender $v \notin \mathcal{S}$, for instance to look like an ordinary user of known contracts, since v can then move those tokens. Such an edge also lets score mass leave $\mathcal{S}$.

With ℓ_{uv} the expected cost to the attacker of one such exposure over one window,

$$C_{\text{exp}} = q \sum_{(u,v)\in E_{\mathcal{S}},\ v\notin\mathcal{S}} \ell_{uv},$$

which is zero for a closed farm.

3. Revocation and detection cost. Legitimate users revoke old allowances from time to time, and a farm whose approvals reappear unchanged in every window might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it. Over q windows,

$$C_{\text{rev}} = q\, c_{\text{rev}}.$$

4. Bought honest edges. The attacker can also pay honest owners to approve its addresses, for instance through an app that asks users for permission. A bought edge (h, a), with $h \in \mathcal{H}$ and $a \in \mathcal{S}$, falls under the same window rule and costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \tag{5.4}$$

where p_h is the payment to the honest owner and ρ_h is a risk premium the owner asks against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{gas}} + C_{\text{exp}} + C_{\text{rev}} + q \sum_{(h,a)\in E_{\mathcal{H}\to\mathcal{S}}} c_{\text{in}}(h, a). \tag{5.5}$$

For a closed farm, $C_{\text{exp}} = 0$ and $E_{\mathcal{H}\to\mathcal{S}}$ is empty, so $C = q\,(|E_{\mathcal{S}}|\, c_{\text{gas}} + c_{\text{rev}})$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves **r** in proportion to the stationary mass at its source node and to its normalized outgoing weight (Langville & Meyer, 2006). Consider first

a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner approves an address in $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j \in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d \sum_{j \in \mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i \in \mathcal{S}} r_i = d \sum_{i \in \mathcal{S}} r_i + |\mathcal{S}|\,(1 - d + d\, r_D)/|V|$, that is,

$$\sum_{i \in \mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|} \cdot \frac{1 - d + d\, r_D}{1 - d}. \tag{5.6}$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \le 1 - \sum_{i \in \mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that holds on any graph:

$$\sum_{i \in \mathcal{S}} r_i \le \frac{|\mathcal{S}|}{(1-d)\,|V| + d\,|\mathcal{S}|}. \tag{5.7}$$

For a farm that is small against $|V|$, the bound is close to $1/(1-d)$ times the population share. On the EndorseRank graph of the matched cohort at T_{obs}, with $6{,}219{,}063$ nodes, the pipeline's own PageRank code gives $r_D = 0.355$, 0.382 and 0.404 at $d = 0.75$, 0.85 and 0.95, so a closed farm on this graph collects 2.06, 3.17 and 8.68 times its population share. A direct check with the same code agrees: a ring of 20 addresses, each approving the next, added to the graph collects 3.17 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which f farm addresses each approve the target t and t approves each of them back. Every farm address a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $c = (1 - d + d\, r_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = c + d\, r_t P_{ta}$ and $r_t = c + d \sum_a r_a$

then give

$$r_t = \frac{c\,(1 + df)}{1 - d^2}.$$

At $d = 0.85$, $c = 7.64 \times 10^{-8}$, which is the common score of every address that receives no allowance. With $f = 10$ (eleven addresses and twenty approvals per window), the target reaches $r_t = 2.62 \times 10^{-6}$ on the graph with the star added, above the median EndorseRank of the cohort contracts but only 8.2×10^{-5} of the highest (0.0318). The score grows linearly in f: each further farm address adds $d\,c/(1 - d^2)$ to r_t and costs two approvals per window. Solved for f, the formula needs about 22 farm addresses to reach the cohort's 90th percentile, about 2,074 its 99th and about 135,895 its highest score. Looking ordinary is therefore cheap and reaching the top expensive: the highest-ranked cohort contract holds about 3% of the graph's stationary mass.

Honest in-edges are therefore not needed to reach the upper part of the ranking, but bought ones add to the gain. When the farm has no out-edges to honest addresses, an edge (h, a) bought from an honest owner h adds, to first order,

$$\Delta \sum_{i \in \mathcal{S}} r_i \approx \frac{d\,r_h\,P_{ha}}{1 - d}$$

to the farm's mass, where P_{ha} is the share of h's row that the new edge takes. The gain depends on this share. A fresh large or unlimited approval bought from h, with $V(z)$ close to one, takes about $\sigma(0)/(o(h) + \sigma(0))$ of h's row; for the median owner in the graph that share is 0.957. The edge an attacker would buy is therefore a large or unlimited approval from an owner whose own approvals are old, small or few, preferably one with a high r_h. Such an approval puts the owner's balance of the token at the attacker's disposal, so the risk premium ρ_h in Equation (5.4) grows with what the owner holds and is close to zero only for an owner that holds none of the token. In the capped allowance layer of C-PR such an approval takes a large share of a row only if it is large against the owner's other allowances (Section 3.6.3). The attack's cost-effectiveness is

$$\eta = \frac{\Delta J(\mathcal{G})}{C}, \tag{5.8}$$

with $\Delta J(\mathcal{G})$ the gain in the objective of Equation (5.3) that the attack edges produce, held over the q windows that C in Equation (5.5) pays for.

Equations (5.6)–(5.8) show that EndorseRank, as defined here, is open to cheap inflation by closed farms. Under uniform teleportation and uniform redistribution of dangling mass, a closed cluster is not held to its population share. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank (Gyöngyi et al., 2004), a cluster outside that set gains mass only through edges from honest owners, which the attacker has to buy at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose addresses send small amounts to one another raises their activeness and so draws restart mass to itself, and EndorseRank with activity restarts inherits the same exposure. The Sybil-adjusted stability proxies (Section 3.9.3) flag repeated transfers from a few addresses but not a farm of fresh addresses, so they cannot replace an adversarial assessment, which we leave to future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Earlier legal work has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024) add a warning of their own: decentralized credit scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. Both EndorseRank and AWP are computed deterministically from on-chain events and a public daily price series with open-source algorithms, so anyone can recalculate them using the open-source software (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with little activity, new ones among them, will score lower whatever their code.

Ethically, these scores should never be used as security ratings, as creditworthiness measures in

the traditional sense, or as evidence that the owners who approve a contract are distinct individuals (Douceur, 2002). They are rank-based indicators of a contract's position in the defined graph topology. If a deployment grants financial services on the strength of score thresholds, it should clearly document the graph definitions behind the scores, the timeframe, and the limitations, above all the exposure to Sybil attacks. A regulatory framework centered on interpretability would prefer this kind of ranking to proprietary machine learning models like those Kotb (2023) reviews, as long as providers make no exaggerated claims about how well they can predict loan repayment rates (Packin & Lev-Aretz, 2024).

Privacy on an open blockchain is limited by the ledger's very nature (Nakamoto, 2008), and computing the reputation scores draws on nothing beyond public data. The main ethical question is therefore how the results are put to use. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study scores contract addresses alone and makes no attempt to link any address to a real-world individual (Section 3.16).

5.8 Threats to validity

The findings reported here should be read against the limitations below, which are grouped under construct, internal, external and adversarial validity.

5.8.1 Construct validity

The threats below concern whether the scores and checks measure what they are read as measuring.

- These ratings are not credit ratings: no default labels from uncollateralized lending enter the analysis.
- Same-window checks rest on local allowance and transfer statistics.
- Holdout construct. A future new approval is still an endorsement event and a future new sender a flow event: both lie outside the score window, yet neither is an independent credit or

trading construct. Being counts of arrivals, they also favor any score that tracks the current count of counterparties (Section 5.1.4).

- Unit of analysis. EndorseRank scores the receiving side of an approval, and every scored address here is a contract (Section 4.1.1). Claims about the reputation of traders rest on the W0 trader run and the comparison of Section 5.1.6, both on small samples.
- Own-construct agreement is partly mechanical. EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for incoming transfers weighted by their value, so each score is expected by construction to agree with its own family. The within-score contrasts of Table 8.6 were computed for this reason; with few ties they compare like with like, but they remain checks of the instruments.
- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with a slope set so that the median amount weighs one half (Section 3.5.1), so the weights record the value of the approvals and transfers an edge carries as well as their number and age. The slopes are choices of the revision, checked only in the same window (Table 4.7). The layers of C-PR add dollar amounts without the bound of V, capping each allowance at U (Section 3.6.3). On both spender holdouts EndorseRank ranks both labels slightly better than the allowance layer walked alone (Tables 4.10 and 4.16), without a paired contrast, so how amounts are weighted may matter for this construct, and neither weighting is shown to be the right one. Such weighting choices propagate into **r** and into the rank correlations (Langville & Meyer, 2006).

5.8.2 Internal validity

The threats below concern whether the comparisons are free of choices that could have produced their results.

- Shared observation window. Every method and proxy is computed over 1 October 2023 to 30 June 2026. Market regimes peculiar to those months (volatility, incentive programs, pro-

tocol changes) may push alignment up or down. The checks on damping and value slopes measure sensitivity to these parameters, but none of them varies the calendar window.

- Cohort selection. The matched cohort is the 15,107 contracts on Arbitrum One that received non-zero approvals of one of the five tokens from at least three owners in the window, and results may not carry over to contracts with fewer approvals or with approvals of other tokens only. Nor do the graphs see every cohort contract (Section 4.1.1): kept as isolated nodes, the contracts outside a graph change no coefficient (Table 4.5), and the sample-definition sensitivity shows how strongly the coefficients react when the evaluation population changes (Table 4.8).
- Baseline fidelity. AWP follows the published method in its edge weights and its restarts (Section 3.6.1). The paper gives no parameter values, so k, t_0 and b are this study's choices (see value weights above). The paper scored Ethereum transactions; here AWP scores transfers of five ERC-20 tokens on Arbitrum One, valued in US dollars. EndorseRank takes AWP's weights, with its own slope for allowances, and restarts uniformly, as fixed in the analysis plan; the variant with AWP's restarts is reported beside it (Table 4.9).
- Order of analysis. The cohorts, scores, labels, statistics and rule A were fixed in the analysis plan of 9 October 2026, before any log of the study was extracted. Rule B with its margin, the two sensitivity analyses and the comparison on trader outcomes were registered later that day, after the W0 results were known and before the study extracted the July–September logs (Section 3.7.2). Every other comparison, the degree contrasts among them, was computed after the relevant labels were known and is reported with intervals; none of them decides a question.
- Revision of the amounts. The amounts were revised on 10 October 2026, after the registered results of all three windows were known (Section 3.5.1). The revised analysis, the main analysis of this thesis, is therefore post hoc, and its verdicts under rules A, B and R1–R3 are a re-analysis, not registered tests. The revision was committed to the public repository before

any revised score was computed, its slopes and cap come from quantiles of the events up to t_1, and the registered analysis reached the same verdicts (Appendix 8.6, Table 8.12).

- Timing environment. All runtimes are wall-clock times from one machine (Appendix 8.1). The standard deviations in Table 4.1 cover only repeated solves of one configuration, and in the damping sweep each graph–damping pair is timed once. Variation between machines is not measured, so the comparison is stated in ratios instead of seconds.

5.8.3 External validity

The threats below concern how far the results carry beyond the data studied.

- Single chain and window. The evidence comes from Arbitrum One alone, from contracts that at least three owners approved on one of five tokens, over 33 months of scored activity and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or lending-centered venues would need cross-protocol validation, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).
- Sample scaling versus alignment scaling. The subsample stages support only a statement about runtime on subgraphs of the matched-cohort graph (Section 4.2.1). Same-window alignment remains anchored to the matched cohort, and the holdout to spenders. The efficiency gains should not be taken to preserve the same τ profile at arbitrary population sizes.
- Ego-network depth. Each graph keeps only the edges incident to the cohort contracts, so longer paths exist only through cohort contracts that both approve and are approved (Section 5.3.2), and the findings about propagation do not carry over to graphs with longer endorsement paths.

5.8.4 Adversarial validity

The threats below concern what an attacker could do to the scores.

- Sybil and wash trading. Rankings may be manipulated through adversarial allowance or trans-

fer patterns (Douceur, 2002); the Sybil-adjusted proxies only partly address this, and the cost model of Section 5.6 is analytical, not an empirical red-team evaluation.

- Strategic approval behavior. Once EndorseRank is live, rational actors may grant allowances strategically to farm rank. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, EndorseRank's strengths are agreement with the allowance checks, a ranking distinct from AWP's, a low cost owed to graph sparsity and, on spender contracts, a ranking of new approval pairs in both windows, a label on which the scores built on transfers alone stay lower. Its weaknesses are a ranking that carries little transfer information, a holdout coefficient below that of the raw in-approve degree it smooths in both windows, exposure to closed farms below the top of its ranking, and nothing to say about contracts that few owners approve or about externally owned accounts, which are not scored (Section 3.2).

The coupled operator was tried at three fixed values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance capped at U. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains went untested, so both answers to RQ5 apply only to that family of methods on this dataset. Deployment remains bound by the considerations of Section 5.7.

The data add four limits. The graphs hold five tokens only (Table 8.4), so a contract that owners approve only on other tokens is outside the cohort. Every price comes from one public service, the daily series of DefiLlama (DefiLlama, n.d.), so an error in it on some date would pass into every amount of that day (Appendix 8.4.2). The account type was read at one block after both label windows, so a contract that destroyed itself before EIP-6780 (Ballet et al., 2023) is left out as an externally owned account (Section 4.1.1). And the trader cohorts are small (Section 4.1.1), so every

trader result has wide intervals.

EndorseRank draws on no off-chain evidence about who controls an address; its only off-chain inputs are the token list and the daily price series. Unlike attestation tooling such as Human Passport, which uses identity checks, biometrics and vouching to establish that an address belongs to a distinct person (Human Passport, n.d.), it gives no guarantee that two highly ranked addresses are controlled by different actors, so it should be read as an ordering of an already admitted population and not as a Sybil defense. Counting only the approvals of owners that pass an attestation gate before computing EndorseRank is a plausible deployment, but the gate's own weaknesses would carry over to it (Bordeianu & Popescu, 2026), and evaluating the combination needs labeled Sybil data that this study did not collect.

These limitations narrow what can be concluded. For this cohort of contracts and time frame, EndorseRank is a cheap, allowance-specific reputation score for spender contracts; it is not a credit scoring method, nor a Sybil-resistant identity mechanism, nor a substitute for protocol-level collateralization (Gudgeon, Perez, et al., 2020). The allowance edge it rests on is worth adding to on-chain reputation, and the approval count is the cheapest way to use it. Chapter 6 restates what has been shown, what that knowledge is worth and what it would take to extend these bounds.